MILITARY SPECIFICATION

BURSTER, PROJECTILE, XM53E2, LOADING ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling and packing of one burster designated as Burster, Projectile, XM53E2.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550	-Ammunition and Special Weapons, General Specification for
MIL-C-45662	-Calibration System Requirements

STANDARDS

MILITARY

MIL-STD-105	-Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109	-Quality Assurance Terms and Definitions
MIL-STD-453	-Inspection, Radiographic
MIL-STD-1168	-Lot Numbering of Ammunition
MIL-STD-1235	-Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

ARMY

8799204	-Carton, Packing, Ammunition, for Charge, Burster, XM53E2.
8799205	-Box, Packing, Ammunition for Charge, Burster, XM53E2.

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9223129 -Projectile, Burster, XM53E2 Assembly
9241806 -Elastomer (Liquid Urethane Rubber)

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material.-Material shall be in accordance with applicable drawings and specifications.

3.1.1 Clean explosive scrap.-Clean explosive scrap produced in regular operations may be remelted and reused. Scrap from floor sweeping, ventilation and suction apparatus (other than from drilling operations specifically controlled to prevent contamination of explosive) shall not be used (see 6.4).

3.2 Burster.-The burster shall comply with all requirements specified on Drawing (dwg.) 9223129 and with all requirements specified in applicable specifications.

3.3 Cavitation

3.3.1 Area from the bottom to two and one half inch above bottom. There shall be no single cavity having a projected length of more than 1/8 inch or a sum total of cavities of more than 1/64 square inch. There shall be no cracks or annular rings in the charge in this area, except that cracks less than 1/32 inch wide shall be disregarded.

3.3.2 Area from top to two and one half inch from bottom of tube. There shall be no single cavity having a projected length of more than 3/8 inch or a sum total of cavities of more than 3/8 square inch and there shall be no more than three transverse cracks over 1/32 inch wide, in this area cracks less than 1/32 inch shall be disregarded.

3.3.3 Porosity.-Porous areas shall be treated as cavities, and shall be subjected to restrictions placed upon cavities, except that 80 percent of the projected length and 80 percent of the projected area shall be considered for acceptance purposes.

3.3.4 Void between charge and base. There shall be no

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void greater than 1/32 of an inch between the charge face and the face of the base.

3.4 Resin.-The resin content shall be 2 percent minimum when tested as specified in 4.3.2.1.

3.5 Workmanship

3.5.1 All parts shall be fabricated and finished in a thorough, workmanlike manner. They shall be clean and free of burrs, sharp edges, unblended radii, cracks, surface defects, chips, dirt, grease, rust and other foreign matter. The cleaning method used shall not be injurious to any parts nor shall the part be contaminated by the cleaning agent. All required markings shall be neat and sharply defined.

3.5.2 Loading of high explosive.-The interior surface of the burster shall be dry when the high explosive is poured into the burster. Solid explosives shall not be present in the burster before loading or be added to the burster during pouring or cooling.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define the terms used herein. The provisions of Specification MIL-A 2550 shall apply.

4.1.1 Submission of product.-At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the con-

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tractual documents, and the certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 Inspection provisions

4.2.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specification, or drawing not affecting safety, performance, interchangeability, or storage as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each lot shall contain:

a. Cyclotol of one type from one manufacturer.

b. TNT of one grade from one manufacturer.

c. Polyurethane elastomer prepolymer from one manufacturer.

d. Glyceride of 12-Hydroxy Stearic acid from one manufacturer.

e. Parts produced by one manufacturer in accordance with same drawing and same drawing revision.

4.2.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.2.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the

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the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

4.2.2.1 Plug, base (see dwg. 9242080 covering a detail of dwg. 9223129).

Categories	Defects	Method Of Inspection
Critical:	None defined.	
Major:	None defined.	
Minor:	AQL 2.50 percent	
201.	Large diameter	Gage
202.	Small diameter, min	Gage
203.	Total length	Gage
204.	Distance to shoulder of large diameter	Gage
205.	Evidence of poor workmanship (see 3.5)	Visual

4.2.2.2 Tube, burster, prior to coating (see dwg. 9223128 covering a detail of dwg. 9223129).

Categories	Defects	Method Of Inspection
Critical:	None defined.	
Major:	None defined.	
Minor:	AQL 1.50 percent	
201.	Total length, min	Gage
202.	Diameter, min	Gage
203.	Wall thickness	Gage
204.	Evidence of poor workmanship (see 3.5)	Visual

4.2.2.3 Tube, burster (see dwg. 9223128 covering a detail of dwg. 9223129).

Categories	Defects	Method Of Inspection
Major:	AQL 0.40 percent	
101.	Interior surface not completely coated	Visual

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102. Foreign matter in tube.....Visual

Minor: None defined.

4.2.2.4 Assembly (see dwg. 9223129).

Categories	Defects	Method Of Inspection
Critical:		
1.	Base plug missing	Visual
2.	Burster tube cracked	Visual
3.	Explosive charge loose in container	Visual
4.	"Up" arrow points in wrong direction	Visual
5.	Closure missing	Visual
Major:	AQL 0.40 percent	
101.	Total length	Gage
102.	Chamber gage	Gage
Minor:	AQL 1.50 percent	
201.	Closure or base plug incompletely sealed	Visual
202.	Evidence of poor workmanship (see 3.5)	Visual
203.	Marking misleading or unidentifiable	Visual

4.2.2.5 Carton prior to sealing (see dwg. 8799204).

Categories	Defects	Method Of Inspection
Critical: None defined.		
Major:	AQL 0.40 percent	
101.	Burster projectile missing	Visual
Minor:	AQL 0.65 percent	
201.	Any packing component missing	Visual

4.2.2.6 Sealed carton (see dwg. 8799204).

Categories	Defects	Method Of Inspection
Critical: None defined.		
Major:	AQL 0.40 percent	
101.	Tape improperly applied	Visual
Minor:	AQL 1.00 percent	
201.	Marking misleading or unidentifiable	Manual
202.	Contents loose upon shaking	Manual

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4.2.2.7 Carton in bag, sealed (see dwg. 8799204).

Categories	Defects	Method Of Inspection
Critical:	None defined.	
Major:	AQL 0.40 percent	
	101. Bag improperly sealed, torn or cut	Visual
Minor:	None defined.	

4.2.2.8 Sealed wood packing box (see dwg. 8799205).

Categories	Defects	Method Of Inspection
Critical:	None defined.	
Major:	AQL 0.40 percent	
	101. Box damaged	Visual
	102. Strapping missing, broken or loose	Visual-Manual
Minor:	AQL 1.50 percent	
	201. Strapping improperly assembled	Visual
	202. Lid improperly assembled	Visual
	203. Markings misleading or unidentifiable	Visual
	204. Contents loose in box	Manual

4.2.3 Testing

4.2.3.1 Cavitation (see 3.3).-Cavitation defects are classified as follows:

Critical defect

Any cavity, crack or annular ring in the area from the bottom to 2 ½ inch above the bottom except cracks less than 1/32 shall be disregarded (see 3.3.1). Any void between charge and base plug in excess of 1/32 inch (see 3.3.4). Cavitation or cracks in excess of twice that permitted by 3.3.2 or 3.3.3.

Major defect

Cavitation or cracks in excess of that permitted by 3.3.2 or 3.3.3.

The sampling plan and procedure for this test for both critical and major defects shall be in accordance with Standard MIL-STD-1235 using a CSP-2 plan having an AQL of 0.25 percent except that when a critical defect is found, 100 percent inspection for i successive defect free units (for both criticals and majors) shall be required immediately instead of continuing f phase of the plan; inspection shall continue at the same frequency until the homogeneity requirement is fulfilled or 100 percent inspection is required. Appropriate l and f values shall be as specified by the procuring activity. The test shall

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be performed as specified in 4.3.1.

4.2.3.2 Composition (see dwg. 9241806) Major defect, -Ten bursters shall be randomly selected from each lot. A sample of composition B-5 shall be taken from an area located one inch from the bottom of each tube, and combined with a sample taken from an area located one inch from the top of the same tube. This procedure shall yield ten composite samples, one from each burster. If any one sample fails to comply with any of the requirements specified the burster lot shall be rejected. The tests shall be performed as specified in 4.3.2.

4.2.4 Inspection Equipment.-The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract (see 6.2).

4.2.4.1 Inspection Equipment Capability and Accuracy.-All inspection equipment and facilities shall be of adequate quality and quantity to perform properly and accurately all the required inspections, measurements and tests. The inspection equipment shall be maintained in satisfactory condition for the continued proper performance of these functions and shall be calibrated in accordance with the provisions of MIL-C-45662.

4.2.4.2 Government Rights to Documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements herein may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.3 Test methods and procedures

4.3.1 Determination of cavities.-The sample bursters shall be subjected to x-ray examination. In all exposures the x-ray beam shall form a 90 degree angle with the longitudinal axis of the projectile. Examination of the x-ray negative shall be made to determine compliance with applicable requirements.

Note.-Radiographic equipment, operations and procedures shall be qualified in accordance with MIL-STD-453. In addition, the sensitivity of x-ray technique shall be 2.0 percent, and the photographic density of the film used shall be no less than 1.25 as determined in accordance with 4.3.1.1 and 4.3.1.2. In the event an original x-ray picture leaves doubt as to acceptability, the burster shall be re-x-rayed at an angle

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90 degrees from that of the original exposure and in the same plane.

4.3.1.1 Check of sensitivity of x-ray technique.-The sensitivity of x-ray technique shall be determined by means of a penetrameter fabricated as specified herein. The penetrameter shall be placed on the side of the burster away from the film (source side). Observation shall be made of the image of the penetrameter on the x-ray film. The sensitivity of x-ray technique shall be considered satisfactory when the image of the penetrameter, including the T2 hole, is sharply defined on the x-ray film. Unless otherwise specified, the penetrameter shall be made of steel having an approximate specific gravity of 7.8. The thickness of the penetrameter shall be based on a combination of both charge and steel thickness and shall be equal to 2.0 percent of the maximum equivalent thickness. The penetrameter shall have a hole, the diameter of which is twice the thickness of the penetrameter. For further details on construction of the penetrameter see MIL-STD-453. The actual thickness may be determined from a radiograph of the metal parts. The penetrameter thickness shall be determined as illustrated in Figure 1 of this specification.

4.3.1.2 Check of photographic density of film.-The photographic density of the film shall be determined on the film in a region of sound charge plus metal corresponding to that portion of the material which is of the greatest combined thickness to be examined on the negative. To judge densities to be used (see 6.3). Comparison of film with standards may be made visually; however, the use of a densitometer is recommended.

4.3.2 Composition

4.3.2.1 Resin.-To an accurately weighed 1 gram (gm.) specimen contained in 250 milliliter (ml.) Erlenmeyer flask 100 ml. of 70 percent by volume acetic acid shall be added. The flask shall be fitted with an air condenser, placed on a steam bath and heated for two hours with frequent shaking. While still warm the insolable residue of the flask shall be transferred to a tared medium or coarse porosity fritted glass crucible. The residue shall be washed with 50 ml. of 70 percent acetic acid, then with 25 ml. of water. The crucible shall be dried to constant weight in a vacuum desiccator or shall be dried in an oven at 70 degrees Centigrade for two hours. The insoluble residue shall be calculated as the percent resin.

4.3.2.2 TNT content.-An accuractely weighed portion of exactly 1.000 gm. of the sample shall be placed in a 100 ml. volumetric flask. About 60 ml. of benzene saturated with RDX shall be added. The flask shall be loosely stoppered and placed on a steam bath for 5 to 10 minutes and occasionally swirled. The flask shall be removed from the steam bath and cooled to room temperature. The flask shall be filled to the mark with saturated benzene and shaken well. The undissolved RDX shall be allowed to settle to the bottom. Using a pipet, 20 ml of the clear supernatant solution shall be transferred to a 500 ml. carbon dioxide (CO sub 2) titration flask. The air shall be swept out using CO sub 2. The flow of CO sub 2 shall be continued through out the determination. Twenty-five ml. of acetic acid and 25 ml. of concentrated hydrochloric acid shall be added to the titration

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flask. The contents shall be stirred by means of a magnetic stirrer for 10 minutes. Exactly 100 ml. of 0.2 normal (N) chromous chloride solution shall be added to the flask and again stirred for 10 minutes. Then drops of a 2 percent aqueous solution of phenosafranine indicator shall be added. The solution shall be titrated with 0.15 N ferric ammonium sulfate solution to a sharp color change from green to deep red. A blank determination shall be run. The percent TNT shall be calculated as follows:

$$\text{Percent TNT} = \frac{6.310\ (N)\ (A - B)}{W}$$

where:

B = ml. of ferric ammonium sulfate used to titrate the sample
A = ml. of ferric ammonium sulfate used to titrate the blank
N = normality of the ferric ammonium sulfate solution
W = weight of the sample

4.3.2.3 RDX content.-An accurately weighed portion of exactly 0.5000 gm. of sample shall be placed in a 100 ml. volumetric flask. About 60 ml. of acid shall be added. The flask shall be heated on a hot plate until the sample is completely dissolved. The flask shall be removed from the hot plate and allowed to cool to room temperature. The flask shall be filled to the mark with acetic acid and shaken well. Using a pipet 20 ml. of contents shall be transferred to a 500 ml. CO sub 2 titration flask. The air shall be swept out using a stream of CO sub 2. The flow of CO sub 2 shall be continued throughout the determination. Twenty five ml. of concentrated hydrochloric acid shall be added to the titration flask. The contents shall be stirred by means of a magnetic stirrer for 10 minutes. Exactly 100 ml. of 0.2 N chromous chloride solution shall be added to the flask and again stirred for 10 minutes. Then 15 drops of a 2 percent aqueous solution of phenosafranine indicator shall be added. The solution shall be titrated with 0.15 N ferric ammonium sulfate solution to a sharp color change from green to deep red. A blank determination shall be run. The percent RDX shall be calculated as follows:

$$\text{Percent RDX} = \frac{6.165\ (N)\ (2\ (C - D) - (A - B))}{W}$$

where:

C = ml. of ferric ammonium sulfate used to titrate the blank
D = ml. of ferric ammonium sulfate used to titrate the sample
A = ml. of ferric ammonium sulfate used to titrate the blank in the TNT determination
B = ml. of ferric ammonium sulfate used to titrate the sample in the TNT determination
N = normality of ferric ammonium sulfate solution
W = weight of sample used in TNT

5. PREPARATION FOR DELIVERY

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5.1 Preservation and packaging

5.1.1 Level C.-The XM53E2 projectile burster shall be preserved and packaged in accordance with dwg. 8799204.

5.2 Packaging

5.2.1 Level C.-The cartons shall be packed in accordance with dwg. 8799205.

5.3 Marking.-The cartons and boxes shall be marked in accordance with dwg. 8799204 or 8799205 as applicable.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

6.2 Inspection Equipment Designs

6.2.1 The contractor shall prepare designs or specify commercial equipment as required for all inspections described in Paragraphs 4.2.2 and 4.2.3.

6.2.2 Contractor designs shall include, as applicable, operating instructions, calibration procedures and maintenance procedures.

6.2.3 Commercial equipment shall be fully described by means of catalogue sheets or the like. This information will include illustrations and engineering data adequate for evaluation. Designs shall be prepared of any special fixture required to be used with commercial equipment.

6.2.4 The specification number and the applicable defect from Section 4 of this specification shall be referenced on each contractor design.

6.2.5 Designs shall be prepared to category and form (per MIL-D-1000) specified in the Contract Data Requirements List (DD Form 1423). The contractor is referred to MIL-HDBK-204 "Inspection tion Equipment Design" for guidance.

6.2.6 Submissions for Approval.-Contractor designs shall be approved by the Government prior to fabricating or procuring the

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equipment. Designs shall be submitted for approval within the time frame and with the distribution specified in the Contract Data Requirements List (DD Form 1423). However, the completion date for design review will be based on the date of the final submission of designs to Picatinny Arsenal, SMUPA-QA-A-A, design review will be accomplished normally within one month after receipt.

6.3 Film density.-"American Standard for Diffuse Transmission Density" published by American Standards Association, 70 East 40th Street, New York, N.Y., has been found satisfactory as a source of information on film densities.

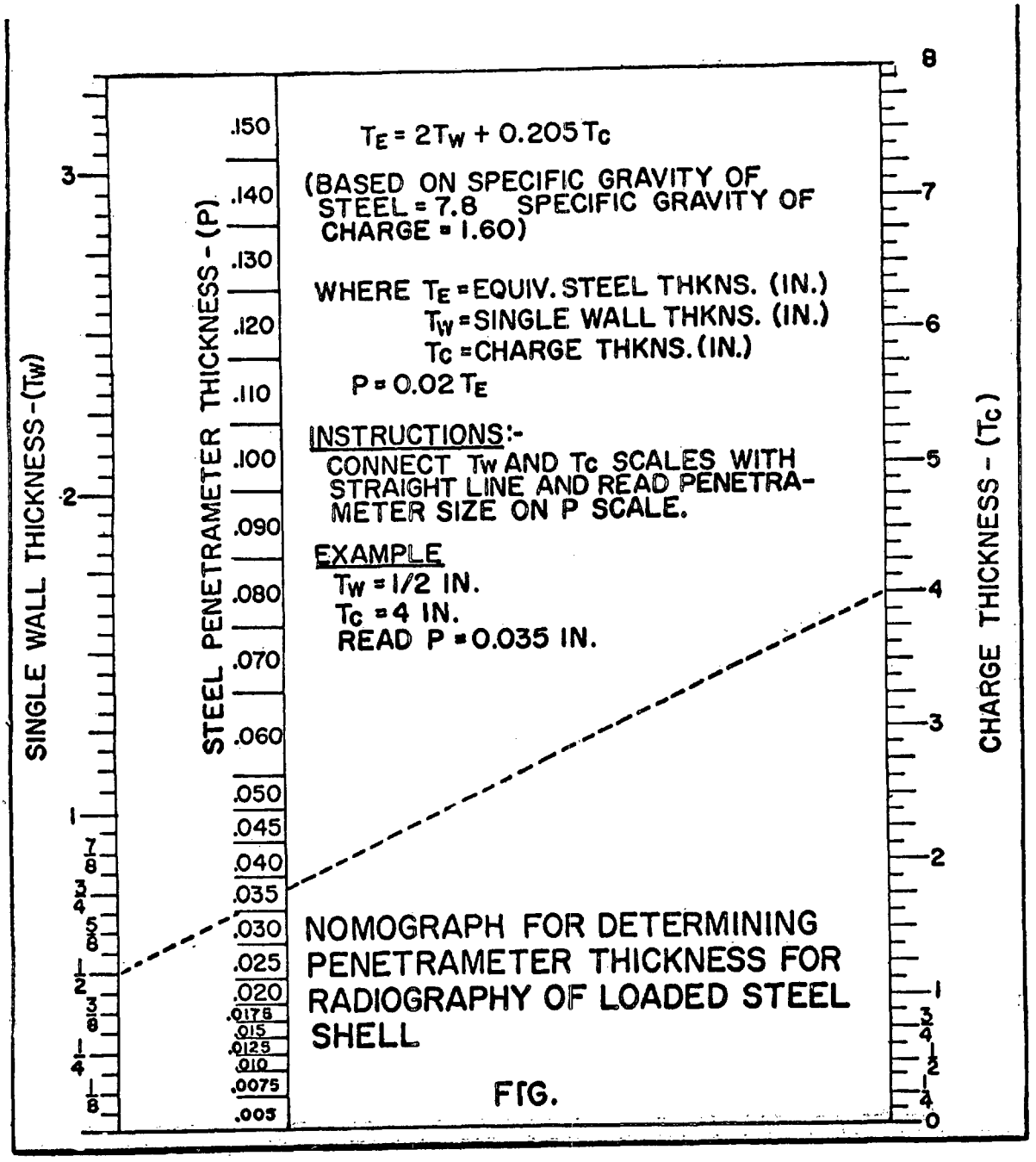
6.4 "Explosive scrap shall not be used in the loading burster."

Custodian:
ARMY-MU

Preparing activity:
ARMY-MU

Project Number: 1315-A-006

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NOMOGRAPH FOR DETERMINING PENETRAMETER THICKNESS FOR RADIOGRAPHY OF LOADED STEEL SHELL

FIG.

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